

Tien Phu Tran

phu2731234@gmail.com | 832 908 2144 | tienphutran.github.io | linkedin.com/in/TienPhuTran

github.com/TienPhuTran | Houston, TX, USA | U.S Citizen

Education

University of Houston , B.S. in Computer Science – GPA: 3.9/4.0	2025
Courseworks: Artificial Intelligence, NLP, Medical Image, Data Science, Database, Web Development.	
Houston Community College , A.S. in Computer Science – GPA: 4.0/4.0	2023

Projects

Sentiment Analysis for Amazon Search Engine	Sep 2025 - Dec 2025
- Designed a semantic search and information retrieval system over 200,000+ Amazon reviews using Python and SpaCy to extract precise aspect-opinion pairs. - Developed an NLP data pipeline with text preprocessing, tokenization, and inverted posting indexes to support fast and scalable search query execution. - Built sentiment polarity metrics using NLTK lexicons and a semantic topic expansion leveraging cosine similarity to improve complex aspect-opinion query analysis. - Improved retrieval quality by using SBERT and BGE ranking models with a custom Rank Penalty algorithm, achieving 86% of peak system accuracy.	
MRI-to-CT Image Translation using GANs (Google Drive)	Sep 2025 - Dec 2025
- Implemented Pix2Pix model in PyTorch using an 8-level U-Net generator for MRI-to-CT conversion and a 70×70 PatchGAN discriminator for adversarial learning between real and synthetic scan. - Collaborated with a team to plan project milestones, coordinate dataset preparation, model development, experimentation, and evaluation following the machine learning development lifecycle. - Developed a preprocessing pipeline using NumPy and Scikit-Image to register and augment 1,250 paired MRI/CT slices from the CHAOS/IXI datasets via spatial alignment, rotations, flips, and intensity normalization. - Trained and optimized the model over 100 epochs on Google Colab using the Adam optimizer and losses. Evaluated performance against a CycleGAN using SSIM, PSNR, and MAE metrics, leveraging OpenCV and Pandas for analysis and visualization between 2 models. - Documented experimental results and presented methodology, model architecture, and performance findings through technical reports and project presentations.	
Document Approval & User Management System (Github)	Jan 2025 - May 2025
- Led a team of 4 students in developing a full-stack Django application for document approval workflows, using audit logging, role-based access, and a dynamic assignment hierarchy for multi-level task delegation. - Planned project milestones, organized task assignments, and encouraged team collaboration by conducting regular progress reviews and code discussions. - Designed a backend pipeline utilizing a LaTeX rendering system and PostgreSQL to systematically store structured data and generate static PDFs. - Collaborated in developing a React frontend using JavaScript, creating a responsive UI for workflow actions, status tracking, and user input management. - Presented system architecture and implementation details to faculty, conveying technical choices and project results.	

Technical Skills

Languages: Python, PowerShell, Bash, SQL, C++ , JavaScript, HTML/CSS

Machine Learning / AI: TensorFlow, PyTorch, OpenCV, Scikit-Image, Hugging Face Transformers, LangChain, CNNs, GANs (Pix2Pix, CycleGAN), Natural Language Processing, Deep Learning

Data & MLOps: Pandas, NumPy, Matplotlib, MLflow, Jupyter Notebook, ETL Pipelines, Regex

Software & Web Dev: Django, React, MySQL, PostgreSQL, REST APIs, LaTeX